

Quantification of arrhythmias using scoring systems: an examination of seven scores in an in vivo model of regional myocardial ischaemia

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ABSTRACT Arrhythmia scores have been used in recent years to facilitate the analysis of arrhythmias, particularly in relation to regional myocardial ischaemia. The recent Lambeth Conventions recommended caution in the use of arrhythmia scores since their use may be misleading. In the present study seven scoring systems were examined in an attempt to validate the use of arrhythmia scores. A strong positive correlation was present between all seven scores. Furthermore, the scores all correlated with the incidences of ventricular fibrillation, ventricular tachycardia, and ventricular premature beats in early myocardial ischaemia. All seven scores successfully detected statistically significant reductions in the incidence of ventricular fibrillation resulting from the administration of two drugs. Some of the scores occasionally showed statistically significant reductions when effects on the raw arrhythmia data were not statistically significant. In this respect, parametric statistical analysis of arrhythmia scores may be a more sensitive method of quantifying arrhythmias than non-parametric analysis of binomially distributed raw data such as the incidence of ventricular fibrillation (in accordance with the power of such tests) indicating that the scores have precision. However, none of the scores incorrectly showed a statistically significant reduction when the raw data expressed a statistically significant or non-significant increase, indicating that the scores have accuracy.

In conclusion, it is possible to design many arrhythmia scores that show changes in arrhythmia severity when more conventional analyses show only non-statistically significant trends. When used in conjunction with raw arrhythmia data, comprehensive drug dose ranges, and appropriate parametric statistical tests, arrhythmia scores facilitate the quantification of arrhythmias. It is recommended that arrhythmia scores should be used only for quantifying group data and model building and not for prognostic purposes in individuals.

We and others have used scoring systems to grade the severity of arrhythmias occurring in animals under experimental conditions of myocardial ischaemia¹⁻⁵ and reperfusion.⁶ By assigning a number to the arrhythmias observed for an animal it is possible to

calculate mean and standard error for a group and thereby conduct parametric statistical testing, provided that the number is part of a Gaussian distributed scale. This approach is more powerful than the non-parametric approach to data analysis, and as a result real effects of interventions may be expected to be found with smaller group sizes.⁷ In practice, estimates of EC₅₀ and ED₅₀ (the concentration or dose of a drug producing a 50% maximum response) for antiarrhythmic interventions may be obtained readily from scoring systems.^{8,9} Equivalent estimates based on binomially distributed variables such as the incidence of ventricular fibrillation are derived from quantal dose-response curves and concentration-

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response curves which are not expected to give estimates of equivalent precision unless group size (n) is increased.⁷

Arrhythmia scores have been used not only in the assessment of the effectiveness of interventions (as an adjunct to the raw arrhythmia data) but also as parameters in mathematical models of arrhythmogenesis¹⁰; the utility of arrhythmia scores in this regard results, again, from their mathematical properties.^{1 10}

The Lambeth Conventions recommended caution in the use of arrhythmia scores.¹¹ In particular, it was recommended that arrhythmia scores should not be emphasised more than the basic arrhythmia data and that arrhythmia scores should not be used as a substitute for the basic data. The score that we have used since 1983 has been applied to examine many interventions for their effects on arrhythmias resulting from regional myocardial ischaemia in conscious or anaesthetised rats and in isolated rat hearts.^{1 8-10 12-19} In the present study, we compared our standard arrhythmia score (referred to as score A) with six new scores in order to examine the validity of such scores.

Our approach was (a) to construct a variety of arrhythmia scores based on different criteria and to establish that they all correlate strongly and positively with the raw arrhythmia data (ventricular tachycardia, ventricular fibrillation, and ventricular premature beats); (b) to establish that these scores correlate strongly and positively with one another; (c) to establish that statistically significant effects of interventions on the raw arrhythmia data may be successfully shown by these scores; and (d) to establish that, although a score might (and should) show statistically significant effects of an intervention at a dose at which effects on the raw arrhythmia data may not reach statistical significance, this property is not misleading in the context of a dose-response study.

Methods

CORONARY OCCLUSION IN CONSCIOUS RATS

Arrhythmia data were derived from an experiment on the effects of (+)- and (-)-verapamil, the results of which have been published in full.⁹ Data from this study were chosen for the present investigation because the drugs produced statistically significant and dose dependent reductions in ventricular fibrillation, ventricular tachycardia, and ventricular premature beats. A discussion of the effects of the drugs on haemodynamic and ECG variables and an interpretation of those results need not be repeated here.

The methods (in brief) were as follows. Male Sprague-Dawley rats were subjected to occlusion of the left main coronary artery while conscious and

unrestrained. Rats ($n=9$ or 18 per group) received saline, (+)-verapamil (0.4, 4, 8, or 12 mg·kg⁻¹), or (-)-verapamil (0.2, 0.6, 2, or 6 mg·kg⁻¹) as slow (10 min) intravenous injections beginning 15 min before coronary occlusion. Drug administration was blinded and randomised, and analyses of the ECG (V3 lead) and other variables⁹ were carried out blind. Single episodes of ventricular tachycardia and ventricular fibrillation lasting longer than 10 s were reverted to sinus rhythm by thump version using the index finger. This practice is common to our studies and is designed to reduce the censoring associated with early mortality.¹⁰

ANALYSIS OF ISCHAEMIA-INDUCED ARRHYTHMIAS

In conscious rats, regional ischaemia produces a distinct biphasic time-severity relation for arrhythmias.¹ Peaks occur at 5-10 min and at 2-3 h after coronary occlusion. Therefore arrhythmias were subdivided into the periods 0-30 min and 0-4 h after occlusion.

Although the present study was carried out before the Lambeth Conventions were drawn up, there are only minor differences in terms of the classification of arrhythmias. Specifically, the number of ventricular premature beats was calculated to include singlets, couplets, triplets, and bigeminy; the Lambeth Conventions recommended that couplets and triplets be analysed as salvos and that bigeminy be analysed separately from ventricular premature beats. Our definitions for ventricular tachycardia and ventricular fibrillation tallied with the recommendations of the Lambeth Conventions.¹¹

The number of episodes of ventricular tachycardia and ventricular fibrillation, the durations of ventricular tachycardia and ventricular fibrillation, and the number of ventricular premature beats were recorded. In our model the durations of ventricular tachycardia and ventricular fibrillation are restricted by thump version (see above). In the absence of this practice, ventricular fibrillation duration becomes impossible to quantify since the duration of lethal ventricular fibrillation has no meaningful value. We have used thump version to provide us with measurable variables, since the durations of ventricular fibrillation and ventricular tachycardia have finite values when thump version is used. The number of ventricular premature beats was expressed as a log₁₀ transform, since this variable is Gaussian distributed (permitting the expression of data as mean(SEM)) whereas the untransformed variable is not.^{1 20}

QUANTIFICATION OF ARRHYTHMIAS

Our standard arrhythmia score (score A) is shown in table 1. Six new scores were also constructed (tables

2-7). All of the scores have three features in common. Firstly, the scores permit the assignation of a single number to each rat from which group means and SEM may be calculated. Secondly, the scores have been designed so that the more severe the arrhythmia, the larger the number. Severity is a subjective concept; we

TABLE 1 *Score A*

0 = 0-49 VPBs
1 = 50-499 VPBs
2 = >499 VPBs and/or 1 episode of spontaneously reverting VT or VF
3 = >1 episode of VT or VF or both (<60 s total combined duration)
4 = VT or VF or both (60-119 s total combined duration)
5 = VT or VF or both (>119 s total combined duration)
6 = fatal VF starting at >15 min after occlusion
7 = fatal VF starting at between 4 min and 14 min 59 s after occlusion
8 = fatal VF starting at between 1 min and 3 min 59 s after occlusion
9 = fatal VF starting <1 min after occlusion

VPB=ventricular premature beats; VT=ventricular tachycardia; VF=ventricular fibrillation.

TABLE 2 *Score B*

0 =<10 VPBs
1 =>10 VPBs
2 =1-5 episodes of VT
3 =>5 episodes of VT and/or any number of episodes of SVF and/or 1 episode of NVF (provided VT and VF total combined duration <40 s)
4 =2-5 episodes of NVF provided VT and VF total combined duration <80 s
5 =>5 episodes of NVF provided VT and VF total combined duration <160 s
6 = VT or VF or both (total combined duration <300 s)
7 = VT or VF or both (total combined duration >300 s)

Abbreviations as in table 1; for VT and VF the prefix S indicates that the arrhythmia reverted to sinus rhythm spontaneously and the prefix N indicates that it did not, and required thump-version.

TABLE 3 *Score C*

0 = No VPBs, VT, or VF
1 = VPBs
2 = 1-5 episodes of VT
3 = >5 episodes of VT or 1 episode of VF or both
4 = 2-5 episodes of VF
5 = >5 episodes of VF

Abbreviations as in table 1.

TABLE 4 *Score D*

0 = <10 VPBs
1 = 10-50 VPBs or 1 episode of VT or both
2 = >50 VPBs or >1 episode of VT or both
3 = SVF or 1 episode of NVF
4 = 2-4 episodes of NVF
5 = >4 episodes of NVF

Abbreviations as in table 2.

have followed the convention that ventricular fibrillation is more severe than ventricular tachycardia, which in turn is more severe than ventricular premature beats. Thirdly, the scores are all isomorphic with arithmetic.

Some scores have a consideration for whether ventricular tachycardia and ventricular fibrillation terminated spontaneously (SVT and SVF respectively) or whether thump version was necessary (that is, that termination was not spontaneous; NVT and NVF respectively). The original score (score A) considers whether ventricular fibrillation was lethal or not; since practice with thump version may eventually lead to 100% successful defibrillation in the conscious rat preparation¹⁰ we did not incorporate a specific consideration for mortality into the new scores. When score A was constructed¹ our premise was that ventricular fibrillation could be subclassified into revertible (by thump version) and non-revertible ventricular fibrillation. Experience has shown us, therefore, that this premise appears to be invalid.¹⁰

Some scores consider ventricular tachycardia and ventricular fibrillation principally in terms of their total durations, whereas some consider ventricular tachycardia and ventricular fibrillation in terms of their

TABLE 5 *Score E*

0 = <20 VPBs
1 = 21-100 VPBs
2 = >100 VPBs or 1-3 episodes of VT or both
3 = >3 episodes of VT
4 = SVF
5 = 1 or 2 episodes of NVF
6 = 3-5 episodes of NVF
7 = >5 episodes of NVF

Abbreviations as in table 2.

TABLE 6 *Score F*

0 = <50 VPBs
1 = 50 or more VPBs
2 = 1-5 episodes of VT
3 = 6 or more episodes of VT
4 = SVF or 1 episode of NVF or both
5 = 2-5 episodes of NVF
6 = >5 episodes of NVF

Abbreviations as in table 2.

TABLE 7 *Score G*

Score = $(\log_{10} \text{VPBs}) + 2(\log_{10} \text{number of episodes of VT}) + 2[(\log_{10} \text{number of episodes of VF}) + (\log_{10} \text{total duration of VF})]$

Abbreviations as in table 1.

number of episodes, and others consider both these aspects. All scores are hierarchical (that is, the score assigned to each rat is the largest possible score) with the exception of score G, which is derived from a simple calculation.

STATISTICAL ANALYSIS

The effects of (+)- and (-)-verapamil were examined by parametric and non-parametric means. The incidences of ventricular tachycardia and ventricular fibrillation were compared using Mainland's contingency tables.²¹ The $\log_{10}$ number of ventricular premature beats and the arrhythmia scores were subjected to analysis of variance and, provided treatment was a significant source of variance, groups were contrasted using Duncan's multiple range test. Analysis was carried out using a UBC computer package²² on an Amdahl mainframe computer. Linear regression correlation was carried out using a MacIntosh desk top computer. $P < 0.05$ was taken as an acceptable level of statistical significance for the examination of the drugs.

Results

EFFECTS OF VERAPAMIL

Both (+)- and (-)-verapamil dose dependently reduced the incidence of ventricular fibrillation, ventricular tachycardia, and the $\log_{10}$ number of ventricular premature beats, as reported previously.⁹ Compared with the control group, the incidence of ventricular fibrillation was reduced ($p < 0.05$) by both of the two highest doses of (+)-verapamil (8 and 12 $\text{mg} \cdot \text{kg}^{-1}$) and by both of the highest doses of (-)-verapamil (2 and 6 $\text{mg} \cdot \text{kg}^{-1}$) during the periods 0-30 min and 0-4 h after coronary occlusion. Score A showed significant reductions ($p < 0.05$) for exactly the same groups, and only those groups when compared with the control group.⁹ Ventricular tachycardia and $\log_{10}$ ventricular premature beats were reduced less effectively by both drugs. $\log_{10}$ ventricular premature beats was statistically significantly reduced only by the highest dose of each drug (not by the three lower doses) and only during the 0-30 min period (not during the 0-4 h period).

ABILITY OF ARRHYTHMIA SCORES TO DETECT EFFECTS OF VERAPAMIL

Figure 1 shows the effects of (+)- and (-)-verapamil on the six new arrhythmia scores and table 8 indicates which effects were statistically significant. In almost every case, statistically significant effects were always shown for the groups in which the incidence of ventricular fibrillation and score A were statistically significantly reduced. The single exception was score C, which failed to show a statistically significant effect

of 2 $\text{mg} \cdot \text{kg}^{-1}$ (-)-verapamil in the 0-4 h period (table 8). Furthermore, in almost every case, each score showed a statistically significant reduction only when there was also a statistically significant reduction in ventricular fibrillation incidence and the value of score A. The single exception was score G, which showed a statistically significant effect of 4 $\text{mg} \cdot \text{kg}^{-1}$ (+)-verapamil during the 0-30 min period.

Table 8 shows that the incidence of ventricular fibrillation was reduced more effectively than that of ventricular tachycardia by both enantiomers of verapamil and that the incidence of ventricular tachycardia was reduced more effectively, in turn, than that of $\log_{10}$ ventricular premature beats, as reported previously.⁹ All seven scores showed statistically significant reductions only in conjunction with statistically significant reductions in the incidence of ventricular tachycardia or $\log_{10}$ ventricular premature beats when the incidence of ventricular fibrillation was also reduced (table 8). Nevertheless, in no instance was $\log_{10}$ number of ventricular premature beats or ventricular tachycardia incidence statistically significantly reduced in the absence of a significant reduction in any of the arrhythmia scores. Thus, in terms of statistically significant effects of the drugs, the scores all corresponded with the incidence of ventricular fibrillation independently of the value of the incidence of ventricular tachycardia only when the incidence of ventricular fibrillation was low, and corresponded with the $\log_{10}$ number of ventricular premature beats only when the incidence of ventricular fibrillation and ventricular tachycardia were low. Since the scores were designed to be dominated by the high numerical values (determined by ventricular fibrillation) this was to be expected.

Despite the dominance of ventricular fibrillation, there was a good positive linear correlation between all seven arrhythmia scores and the incidence of ventricular tachycardia and $\log_{10}$ ventricular premature beats as well as with the incidence of ventricular fibrillation (table 9).

DIFFERENCE BETWEEN SCORES

Table 9 and fig 2 show that all seven scores correlated strongly and positively with one another. The standard errors of the mean were similar for each score (fig 2). The largest values were found with score G (0-4 h data) for which the mean of the SEM expressed as a percentage of each mean was 29(9)%. The smallest values were found for score D (0-4 h data) for which the mean of the SEM was 11(2)%. Analysis of variance showed no significant variance inhomogeneity between groups for any score and no differences between the means of the SEM.

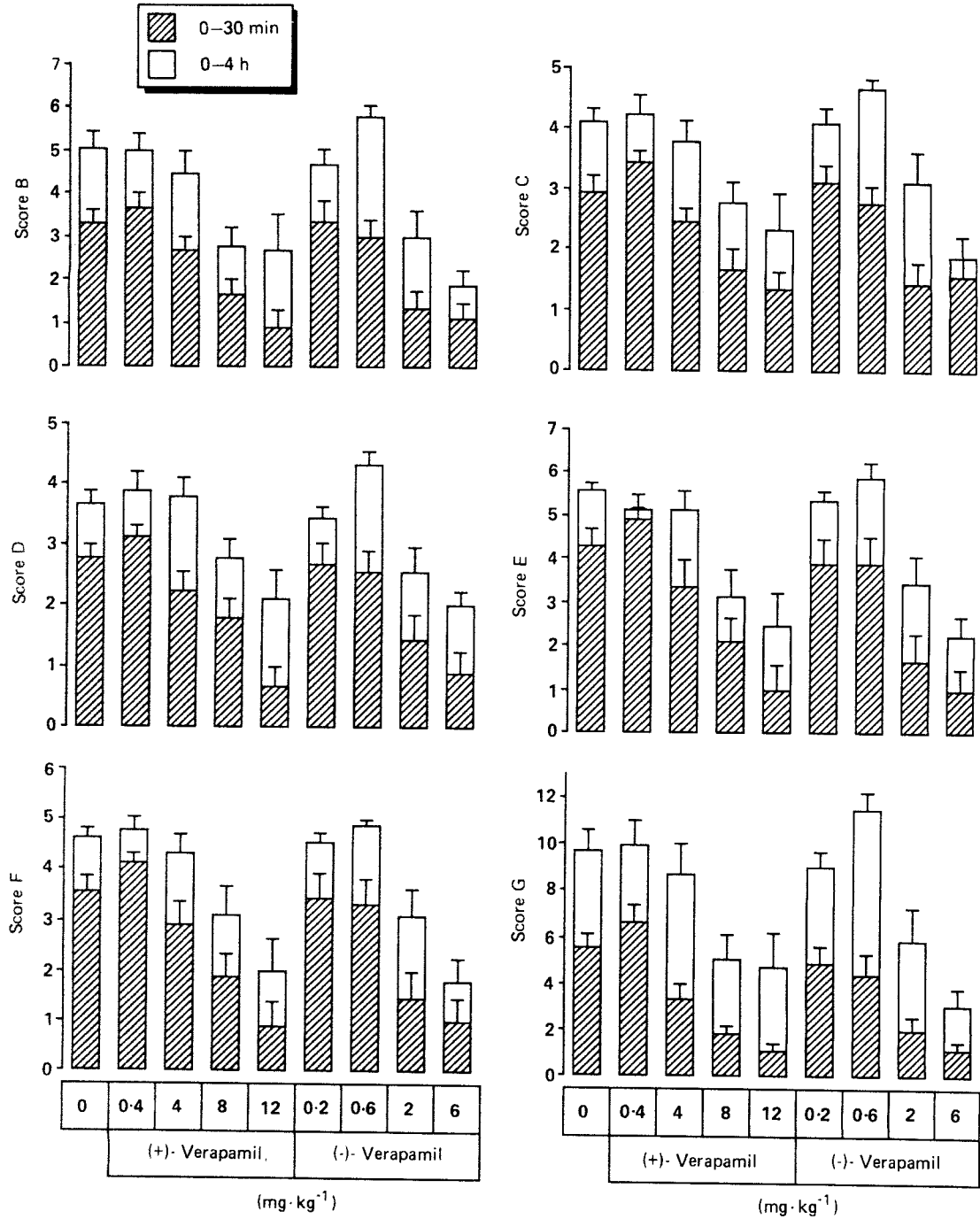


FIG 1 Effects of the optical enantiomers of verapamil on the six new arrhythmia scores. Values are given for the 0-30 min period and the 0-4 h period after coronary occlusion. Values are mean(SEM).

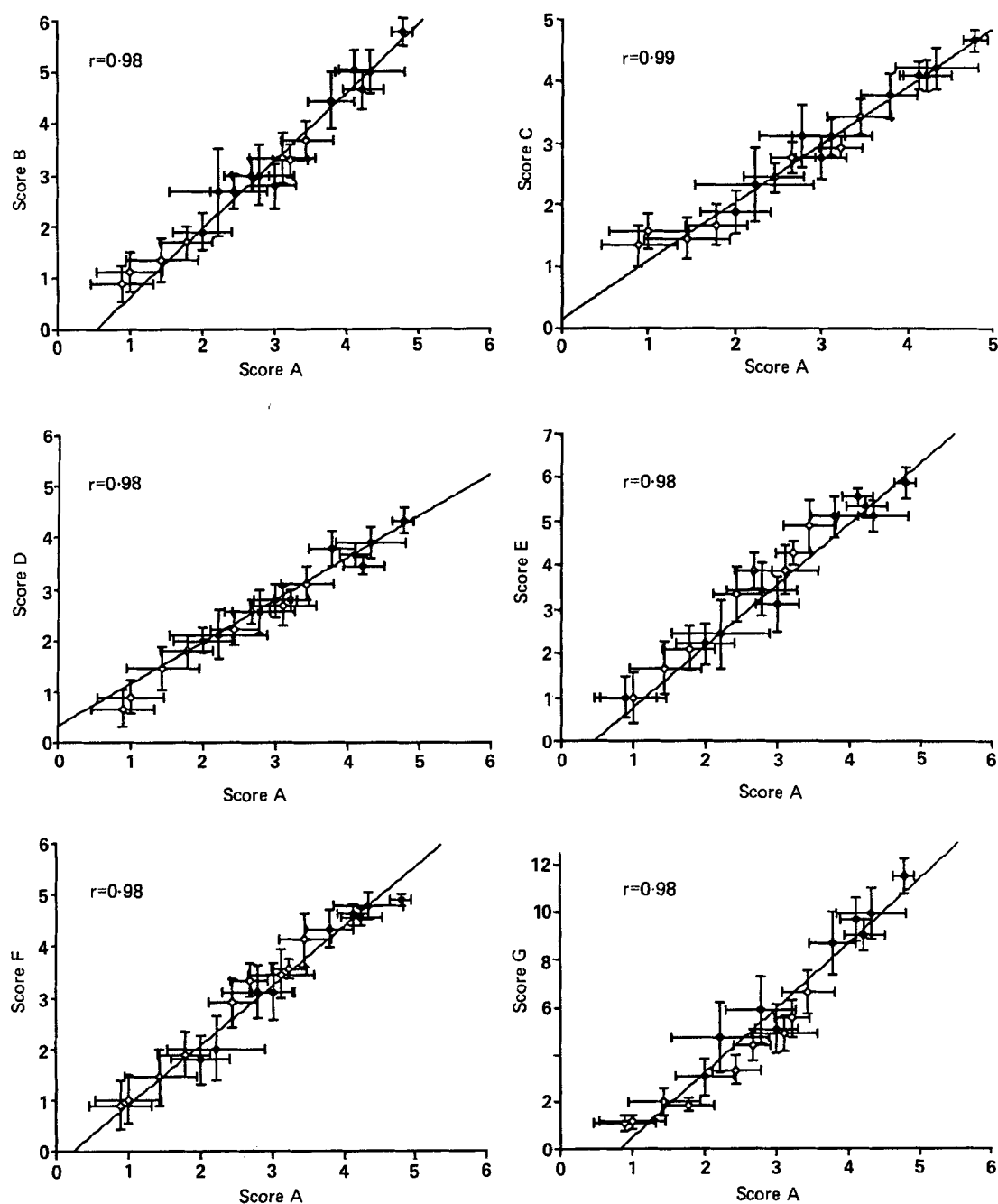


FIG 2 Linear correlations between the six new arrhythmia scores (scores B-G) and the old score (score A). Values are mean(SEM) for the 0-30 min period after coronary occlusion (open symbols) and the 0-4 h period (closed symbols). Regression lines were all significantly different from zero ($p<0.001$).

TABLE 8 Ability of arrhythmia scores (A-G) to detect statistically significant effects of (+)- and (-)-verapamil compared with the control group

Comparison with verapamil dose (mg)	0-30 min data										0-4 h data									
	A	B	C	D	E	F	G	%VF	%VT	Log ₁₀ VPB	A	B	C	D	E	F	G	%VF	%VT	Log ₁₀ VPB
C vs 0.4 (+)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C vs 4 (+)	0	0	0	0	0	0	*	0	0	0	0	0	0	0	0	0	0	0	0	0
C vs 8 (+)	*	*	*	*	*	*	*	*	*	0	*	*	*	*	*	*	*	*	0	0
C vs 12 (+)	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	0
C vs 0.2 (-)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C vs 0.6 (-)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C vs 2 (-)	*	*	*	*	*	*	*	*	*	0	*	*	0	*	*	*	*	*	0	0
C vs 6 (-)	*	*	*	*	*	*	*	*	*	0	*	*	*	*	*	*	*	*	0	0

* $p < 0.05$ for the comparison indicated. Comparisons were conducted using Duncan's test for the arrhythmia and VPB scores and Mainland's contingency tables for %VT and %VF. 0=not statistically significant. Abbreviations as in table 1 (C=controls).

TABLE 9 Correlations between arrhythmias and arrhythmia scores

	A	B	C	D	E	F	G	%VF	%VT	Log ₁₀ VPB
A	—	0.98**	0.99**	0.98**	0.98**	0.98**	0.98**	0.96**	0.9**	0.95**
B	0.98**	—	0.99**	0.98**	0.98**	0.98**	0.97**	0.96**	0.81**	0.94**
C	0.99**	0.99**	—	0.96**	0.98**	0.98**	0.98**	0.98**	0.88**	0.92**
D	0.98**	0.98**	0.96**	—	0.99**	0.99**	0.97**	0.92**	0.89**	0.97**
E	0.98**	0.98**	0.98**	0.99**	—	1**	0.98**	0.99**	0.88*	0.93**
F	0.98**	0.98**	0.98**	0.99**	1**	—	0.98**	0.98**	0.93*	0.94**
G	0.98**	0.97**	0.98**	0.97**	0.98**	0.98**	—	0.97**	0.82*	0.94**
%VF	0.96**	0.96**	0.98**	0.92**	0.99**	0.98**	0.97**	—	0.87*	0.9**
%VT	0.9**	0.81**	0.88**	0.89**	0.88**	0.93**	0.82**	0.87*	—	0.97*
Log ₁₀ VPB	0.95**	0.94**	0.92**	0.97**	0.93**	0.94**	0.94**	0.9**	0.82*	—

Values are correlation coefficients (r). ** $p < 0.001$, * $p < 0.01$. Abbreviations as in table 1.

TABLE 10 Agreement between arrhythmia scores and incidence of ventricular fibrillation in terms of statistically significant differences between groups

Score	0-30 min data				0-4 h data			
	Correct	Excess	Deficit	Incorrect	Correct	Excess	Deficit	Incorrect
A	31/36	5/36	0/36	0/36	30/36	5/36	1/36	0/36
B	29/36	7/36	0/36	0/36	30/36	6/36	0/36	0/36
C	29/36	7/36	0/36	0/36	30/36	4/36	2/36	0/36
D	30/36	6/36	0/36	0/36	29/36	5/36	2/36	0/36
E	29/36	7/36	0/36	0/36	30/36	6/36	0/36	0/36
F	29/36	7/36	0/36	0/36	29/36	7/36	0/36	0/36
G	27/36	9/36	0/36	0/36	32/36	4/36	0/36	0/36

There were 36 (8+7+6 . . . +1) possible comparisons between the 9 groups per time interval. If a statistically significant difference in VF incidence tallied with a statistically significant reduction in a score, then the score was said to be correct. If the score showed a statistically significant difference in the absence of a statistically significant reduction in VF incidence, then the score was said to be in excess. If the score showed no statistically significant difference despite a statistically significant reduction in VF incidence then the score was said to be in deficit. If a score showed a statistically significant difference between 2 groups when there was either no difference in VF incidence or the group with the lower score had the higher VF incidence, then an incorrect result was recorded. All events are shown as an incidence out of the total 36 comparisons per time interval.

Table 8 shows data for eight comparisons (drugs versus controls). However, there were a total of 36 possible comparisons between the nine groups (drugs versus drugs and controls) for both the 0-30 min and 0-4 h periods. Table 10 summarises these comparisons in terms of the agreement between the scores and

incidence of ventricular fibrillation. For each comparison, a score was recorded as being correct when the presence or absence of a statistically significant difference in incidence of ventricular fibrillation tallied with the presence or absence of a statistically significant difference in arrhythmia

scores. A correct result occurred 27-32 times out of 36 for the seven arrhythmia scores. In no instance was a score found to give an incorrect result (defined as a statistically significant reduction in the score for a comparison in which the incidence of ventricular fibrillation was increased or remained unchanged).

We examined in detail how well the scores related to the incidence of ventricular fibrillation, ventricular tachycardia, and $\log_{10}$ number of ventricular premature beats. It is important also to examine how well ventricular tachycardia and $\log_{10}$ number of ventricular premature beats relate to ventricular fibrillation in order to give some perspective to the assessment of the scores. We therefore compared ventricular tachycardia and $\log_{10}$ ventricular premature beats with ventricular fibrillation in the same manner that was used to examine the relation between the scores and ventricular fibrillation (as shown in table 10). This analysis showed that there was a much weaker degree of agreement between the incidence of ventricular fibrillation and both the incidence of ventricular tachycardia and $\log_{10}$ number of ventricular premature beats than between the incidence of ventricular fibrillation and the arrhythmia scores. For the 0-4 h period, the incidence of ventricular tachycardia was correct only 21 times out of 36 and $\log_{10}$ the number of ventricular premature beats was correct only 22 times out of 36. Thus all seven arrhythmia scores tallied considerably more closely with the most severe arrhythmia (ventricular fibrillation) than did either the incidence of ventricular tachycardia or $\log_{10}$ number of ventricular premature beats.

Discussion

MATHEMATICAL PROPERTIES OF ARRHYTHMIA SCORES

Our aim, when we developed arrhythmia scores, was to facilitate the quantification of arrhythmias. It is desirable then that arrhythmia scores should possess appropriate mathematical properties. Our original score (score A) was found to be Gaussian distributed in a review of control data.¹ Consideration of variance (i.e. SEM) and the close correlation between all seven scores suggests that the same is true for the six new scores, although the number of control animals in the present studies was too small to permit distribution analysis. If the scores are Gaussian distributed, as they appear to be, then they are suitable for parametric statistical analysis. There are three other scoring systems currently in use for examining arrhythmias in early ischaemia.²⁻⁵ We did not include them in the present investigation since they are not Gaussian distributed, and all three are therefore unsuitable for parametric statistical analysis.

RELATION BETWEEN ARRHYTHMIA SCORES AND RAW DATA

All the arrhythmia scores correlated strongly and positively with one another and with the raw arrhythmia data. However, all seven scores were more closely related to ventricular fibrillation than to ventricular tachycardia or ventricular premature beats. The reasons for this are (a) the scores were all designed to be more sensitive to the most severe arrhythmia (ventricular fibrillation) and (b) the incidence of ventricular fibrillation was not isologous with that of ventricular tachycardia or $\log_{10}$ number of ventricular premature beats, in that the incidence of ventricular fibrillation was more responsive to the interventions than were the other arrhythmias. The scores were related only to ventricular tachycardia and $\log_{10}$ ventricular premature beats when the incidence of ventricular fibrillation was low. Clearly, an intervention that can reduce ventricular premature beats or ventricular tachycardia or both without reducing ventricular fibrillation would not cause a reduction in any of the scores described here. If it were desired to search for a drug which reduced ventricular tachycardia or ventricular premature beats or both without reducing ventricular fibrillation, then it would not be appropriate to use any of these scores. However, such a drug would not appear to be particularly useful in early ischaemia. Therefore, although there is a built in bias of the scores towards ventricular fibrillation, we do not consider this to be a significant disadvantage of the scores.

PRECISION AND ACCURACY

The Lambeth Conventions recommended that arrhythmia scores should not be used as a substitute for using a larger group size.¹¹ We would concur with this conclusion. Nevertheless, the usefulness of an arrhythmia score of the type discussed here lies in the fact that for any group size the superior precision compared with binomially distributed data (such as the incidence of ventricular fibrillation) allows the effects of interventions to reach statistical significance when real reductions in binomially distributed variables may fail to reach statistical significance. The question that arises is, which system of analysis provides the most accurate information? On the one hand, it is conceivable that a statistically significant reduction in arrhythmia score may be misleading in the absence of a statistically significant reduction in, for example, the incidence of ventricular fibrillation. On the other hand, it may be the case that an intervention at a low dose causes a real reduction in the incidence of ventricular fibrillation which is, nevertheless, not statistically significant owing to the poor precision of non-parametric tests. In the latter case, an arrhythmia score may show the true effect of the intervention at a

lower dose than can be shown (in statistical terms) by non-parametric tests on the raw data.

The latter point is amply illustrated by the present data set. In table 8 it was shown that, compared with the control group, the highest two doses of both drugs reduced the incidence of ventricular fibrillation significantly. In conjunction with this there was a statistically significant reduction in the value of the seven arrhythmia scores in all but one of 56 comparisons. In contrast, the 2 mg·kg⁻¹ dose of (–)-verapamil and the 8 mg·kg⁻¹ dose of (+)-verapamil produced no statistically significant reductions in the incidence of ventricular fibrillation when compared with the low dose groups given 0.4 mg·kg⁻¹ (+)-verapamil or 0.2 mg·kg⁻¹ (–)-verapamil. Examination of the raw ventricular fibrillation data shows that the incidence of ventricular fibrillation in the 0–30 min period was 6 out of 9 and 8 out of 9 in the 0.2 mg·kg⁻¹ (–)-verapamil group and the 0.4 mg·kg⁻¹ (+)-verapamil group respectively and 2 out of 9 and 2 out of 9 in the 2 mg·kg⁻¹ (–)-verapamil group and the 8 mg·kg⁻¹ (+)-verapamil group respectively.⁹ Commonsense might suggest that these incidences are different. However, commonsense is not held to be an acceptable method of analysing data (although see 23). The arrhythmia scores, by virtue of their superior precision over the binomially distributed variable, incidence of ventricular fibrillation, were capable of showing statistically significant differences between these groups.

PROGNOSIS VS ANALYSIS OF GROUP DATA AND MODEL BUILDING

The scoring systems described here should not be confused, conceptually, with the Lown grading system.²⁴ Our scores summarise ischaemia induced arrhythmias in random samples of the population, whereas the Lown grading system classifies the severity of arrhythmias in individuals. The practice of using a grading system in an individual for prognostic purposes is no longer recommended²⁵ owing to the poor correlation between so called warning arrhythmias and mortality.^{26–29} It must be stressed that we do not suggest that arrhythmia scores be used for prognostic purposes in individuals but only for quantification of group data and model building. The latter application may be potentially the more useful. For example, score A has been successfully applied to the analysis of the relation between the volume of the ischaemic region and arrhythmias.^{1–10} On the basis of this analysis it has been proposed that assessment of the influence of an intervention on the relation between the volume of the ischaemic region and score A may facilitate the assessment of the mechanism of action of the intervention.¹⁰ Analysis of this type is almost impossible if a binomially distributed variable (such as the incidence of ventricular fibrillation) is used.

CONCLUSIONS

We have examined seven arrhythmia scores for use in the quantification of arrhythmias resulting from early regional myocardial ischaemia. All the scores were highly effective in detecting the statistically significant reductions in ischaemia induced ventricular arrhythmias produced by (+)- and (–)-verapamil. The present study shows that the use of arrhythmia scores to facilitate the quantitative analysis of arrhythmias is a valid approach. However, the information that such scores provides must be interpreted cautiously. It is clear that antiarrhythmic trends can be stamped with the seal of statistical significance by the use of arrhythmia scores when the nature of the raw data itself cannot be analysed with sufficient precision to be stamped with the same seal of statistical approval. The question that cannot be given a definite answer on the basis of the present analysis is whether or not enhanced precision is accompanied by enhanced accuracy. However, in the case of score A, we can argue in its defence that it has never shown an intervention to be antiarrhythmic in the context of a dose-response study when the raw data do not support such a conclusion.¹⁰

The Lambeth Conventions recommended that drugs be examined for their antiarrhythmic effects over a range of doses.¹¹ In order to preclude false claims on the effectiveness of a drug, arrhythmia scores should be considered only in relation to dose-response data (or the equivalent, in non-drug studies) and in conjunction with the raw arrhythmia data. In this manner, analysis of the overall effect of a drug will be dependent on neither the outcome of a single dose nor a single system of analysis. Finally, we recommend that arrhythmia scores should be used only for quantification of group data and model building and not for prognostic purposes in individuals.

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References

- 1 Johnston KM, MacLeod BA, Walker MJA. Responses to ligation of coronary artery in conscious rats and the actions of antiarrhythmics. *Can J Physiol Pharmacol* 1983;**61**:1340–53.
- 2 Martinez TT, Crampton JM. Prostaglandins show marked differences when tested against early ischemic dysrhythmias in rat. *Life Sci* 1981;**29**:1449–56.
- 3 Mueller B, Schneider J, Hennies HH, Flohe L. Cardioprotective action of the new stable epoprostenol analogue, CG 4203, in rat models of cardiac hypoxia and ischemia. *Arzneimittel Forschung* 1984;**34**:1506–9.
- 4 McLennan PL, Abeywardena MY, Charnock JS. Influence of dietary lipids on arrhythmias and infarction after coronary

- artery ligation in rats. *Can J Physiol Pharmacol* 1985;**63**:1411–22.
- 5 Daugherty A, Frayne KN, Redfern WS, Woodward B. The role of catecholamines in the production of ischaemia-induced ventricular arrhythmias in the rat in vivo and in vitro. *Br J Pharmacol* 1986;**87**:265–78.
- 6 Lubbe WF, Holland RK, Gilchrist AI, Pybus J. Permissive action of potassium on arrhythmogenesis in the rat heart. *J Mol Cell Cardiol* 1986;**18**(suppl 4):37–41.
- 7 Diem K, ed. *Documenta Geigy scientific tables (6th ed)*. Manchester: Geigy Pharmaceuticals: ■156–8.
- 8 Curtis MJ, MacLeod BA, Walker MJA. Antiarrhythmic actions of verapamil against ischaemic arrhythmias in the rat. *Br J Pharmacol* 1984;**83**:373–85.
- 9 Curtis MJ, Walker MJA. The mechanism of action of the optical enantiomers of verapamil against ischaemia-induced arrhythmias in the conscious rat. *Br J Pharmacol* 1986;**89**:137–47.
- 10 Curtis MJ, MacLeod BA, Walker MJA. Models for the study of arrhythmias in myocardial ischaemia and infarction: the use of the rat. *J Mol Cell Cardiol* 1987;**19**:399–419.
- 11 Walker MJA, Curtis MJ, Hearse DJ, *et al*. The Lambeth conventions: guidelines for the study of arrhythmias in ischaemia, infarction, and reperfusion. *Cardiovasc Res* 1988;**22**:00–00.
- 12 Au TLS, Collins GA, Harvie CJ, Walker MJA. The actions of prostaglandin E₂, propranolol, and nitroglycerine with halothane, pethidine or pentobarbitone anaesthesia on arrhythmias and other responses to ligation of a coronary artery in rats. *Br J Pharmacol* 1983;**79**:929–37.
- 13 Botting JH, Johnston KM, MacLeod BA, Walker MJA. The effect of modification of sympathetic activity on responses to ligation of a coronary artery in the conscious rat. *Br J Pharmacol* 1983;**79**:265–71.
- 14 Curtis MJ, Johnston KM, MacLeod BA, Walker MJA. The actions of felodipine on arrhythmias and other responses to myocardial ischaemia in conscious rats. *Eur J Pharmacol* 1985;**117**:169–78.
- 15 Curtis MJ, MacLeod BA, Walker MJA. The effects of ablations in the central nervous system on arrhythmias induced by coronary occlusion in the rat. *Br J Pharmacol* 1985;**86**:663–70.
- 16 Curtis MJ, Walker MJA, Yuswack T. Actions of the verapamil analogues, anipamil and ronipamil, against ischaemia-induced arrhythmias in conscious rats. *Br J Pharmacol* 1986;**88**:355–66.
- 17 Jang TL, MacLeod BA, Walker MJA. Effects of halogenated hydrocarbon anesthetics on responses to ligation of a coronary artery in chronically prepared rats. *Anesthesiology* 1983;**59**:309–15.
- 18 Johnston KM, MacLeod BA, Walker MJA. Effects of aspirin and prostacyclin on arrhythmias resulting from coronary artery ligation and on infarct size. *Br J Pharmacol* 1983;**78**:29–37.
- 19 MacLeod BA, Augereau P, Walker MJA. Effects of halothane anesthesia compared with fentanyl anesthesia and no anesthesia during coronary ligation in rats. *Anesthesiology* 1983;**58**:44–52.
- 20 Mullane KM, Read N, Salmon JA, Moncada S. Role of leucocytes in acute myocardial infarction in anaesthetised dogs: relationship to myocardial salvage by anti-inflammatory drugs. *J Pharmacol Exp Ther* 1984;**228**:510–22.
- 21 Mainland D, Herrera L, Sutcliffe MI. *Statistical tables for use with binomial samples – contingency tests, confidence limits and sample size estimates*. New York: Department of Medical Statistics Publications, New York Medical College of Medicine, 1956.
- 22 Gregg M, Osterlin D. *UBC Anovar*. Vancouver: UBC Computer Centre Publications, 1977.
- 23 Kitchen I. Statistics and pharmacology: the bloody obvious test. *Trends Pharm Sci* 1987;**8**:252–3.
- 24 Lown B, Wolf M. Approaches to sudden death from coronary heart disease. *Circulation* 1971;**44**:130–42.
- 25 Bigger JT Jr, Weld FM. Analysis of prognostic significances of ventricular arrhythmias after myocardial infarction. Shortcomings of Lown grading system. *Br Heart J* 1981;**45**:717–24.
- 26 Campbell RWF, Murray A, Julian DG. Ventricular arrhythmias in the first 12 hours of acute myocardial infarction. Natural history study. *Br Heart J* 1981;**46**:351–7.
- 27 El-Sherif N, Myerburg RJ, Scherlag BJ, *et al*. Electrocardiographic antecedents of primary ventricular fibrillation. *Br Heart J* 1976;**38**:415–22.
- 28 Kennedy HL, Whitlock JA, Sprague MK, Kennedy LJ, Buckingham TA, Goldberg RJ. Long-term follow up of asymptomatic healthy subjects with frequent and complex ventricular ectopy. *N Eng J Med* 1985;**213**:193–7.
- 29 Rabkin SW, Mathewson FAL, Tate RB. The electrocardiogram in apparently healthy men and the risk of sudden death. *Br Heart J* 1982;**47**:546–52.